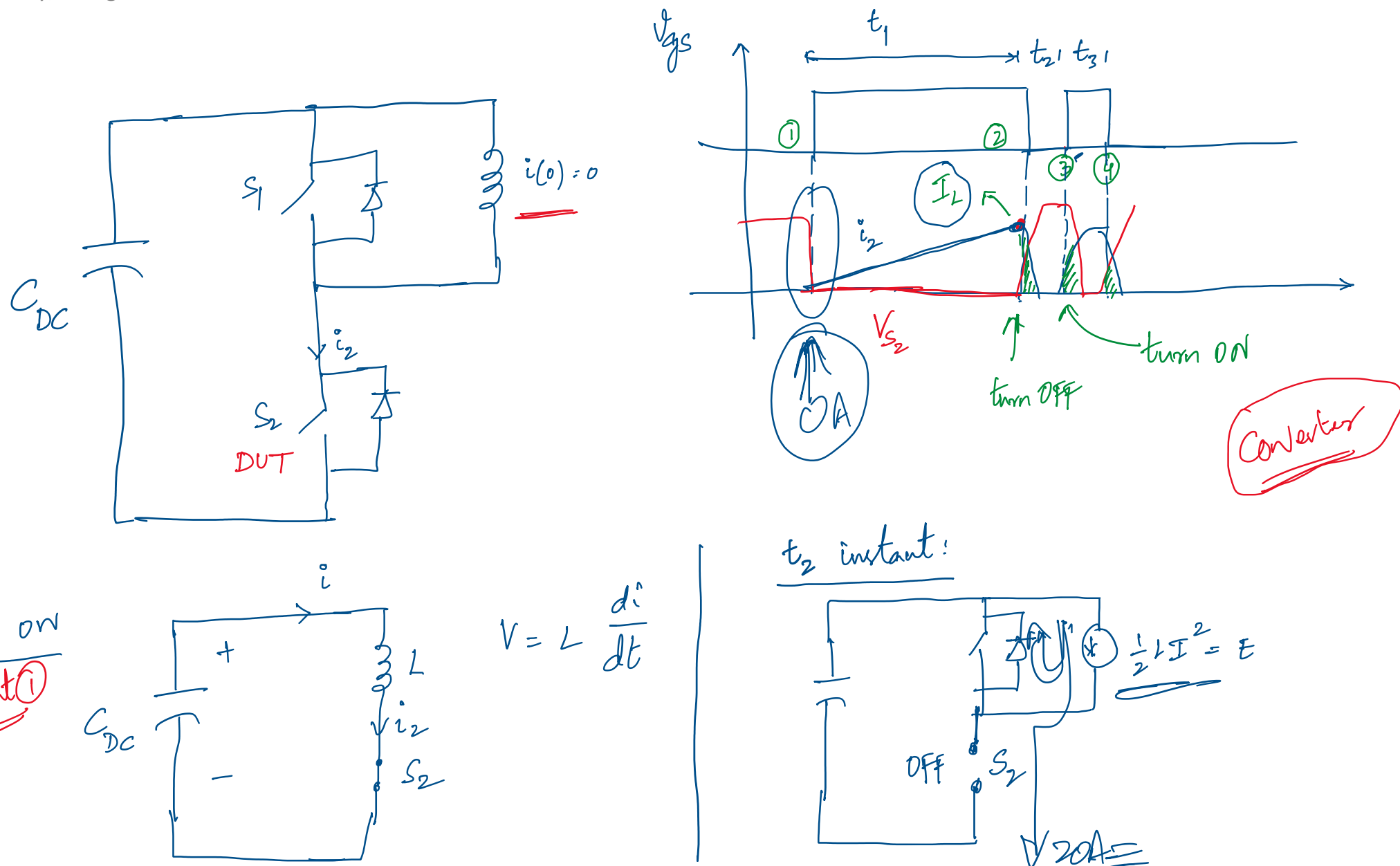


Intra to DPT :-

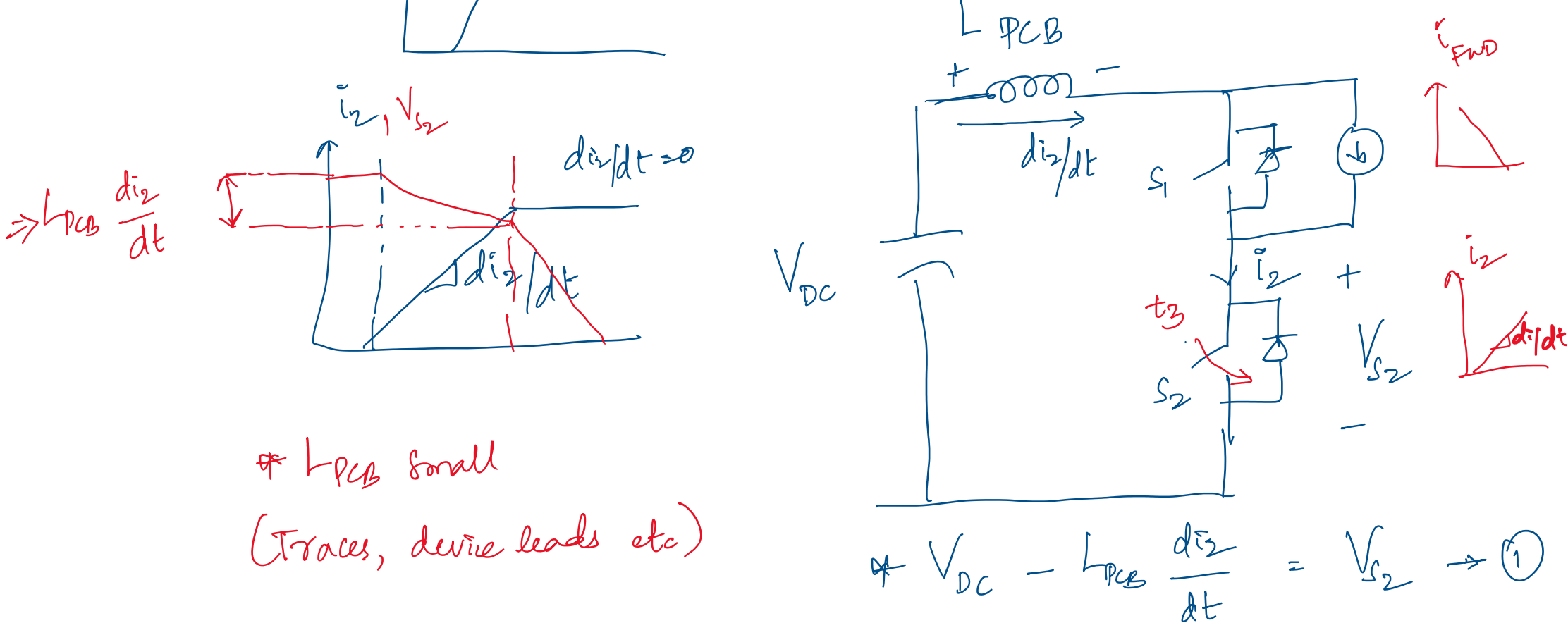
Monday, 7 August 2023 8:48 PM



→ Voltage overshoot during turn OFF, $E_{sw,off}$, t_{fall} , t_{rise} .

→ turn ON: i_L V_{S2}

$E_{sw,ON}$, t_{fall} , t_{rise} .



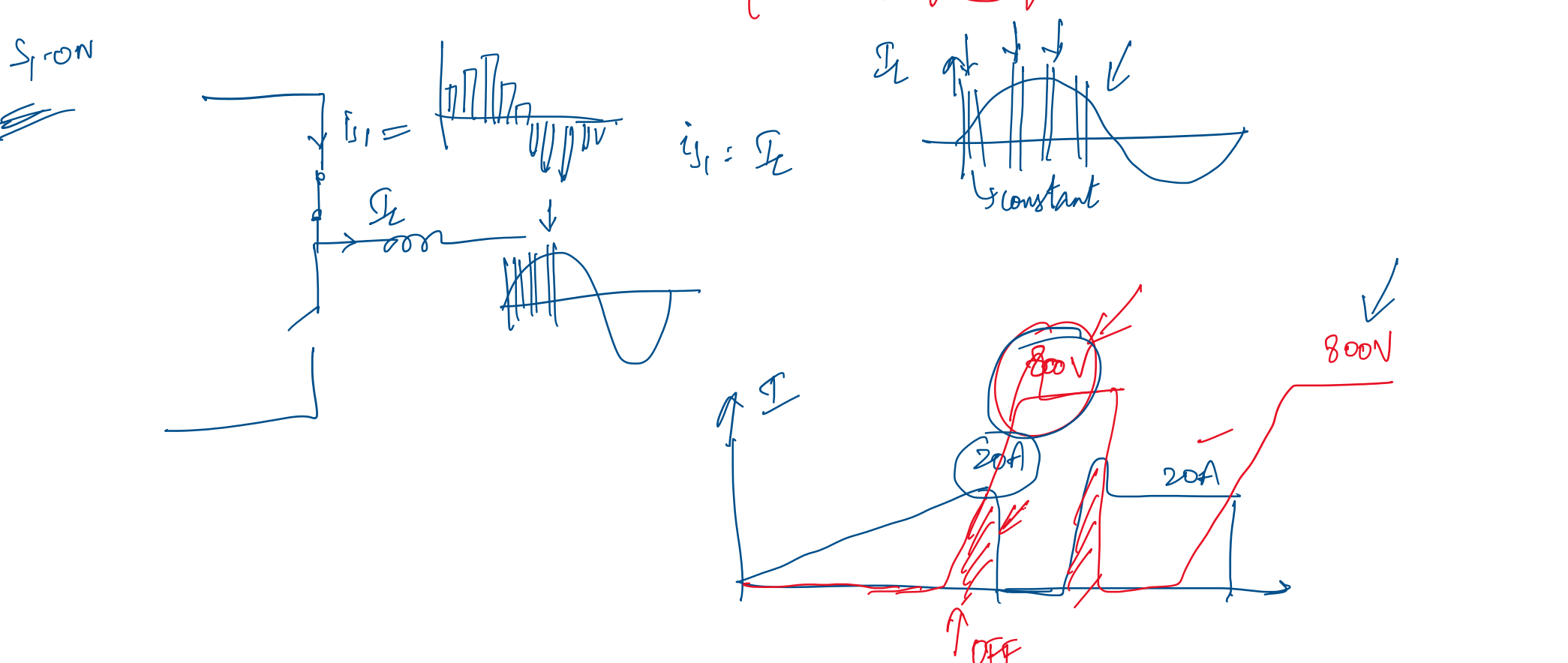
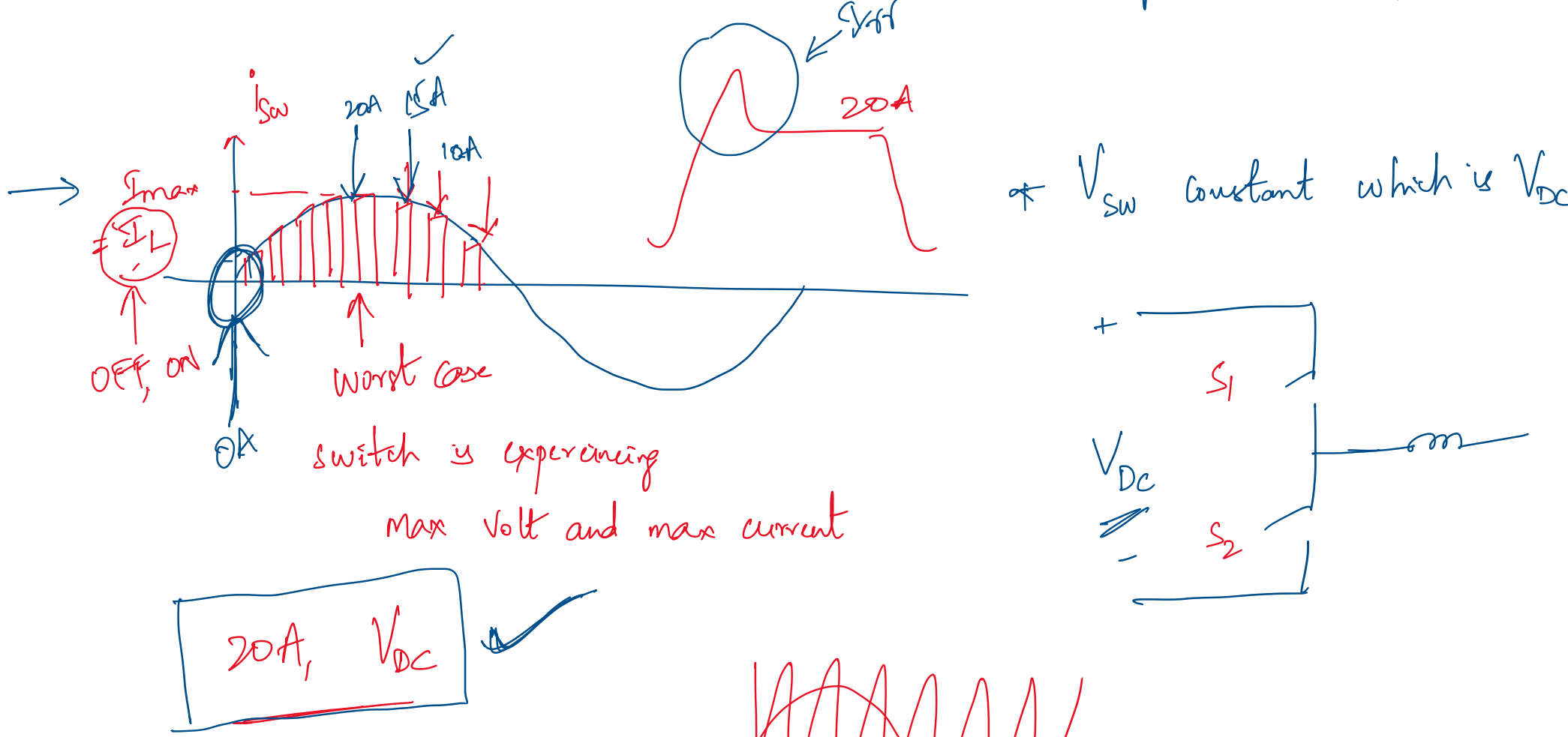
L_{PCB} how to calculate:

→ From turn on charac. (i) $\frac{di_L}{dt}$ is known (measure using cursors in DSO)

(ii) V_{dip} can be measured

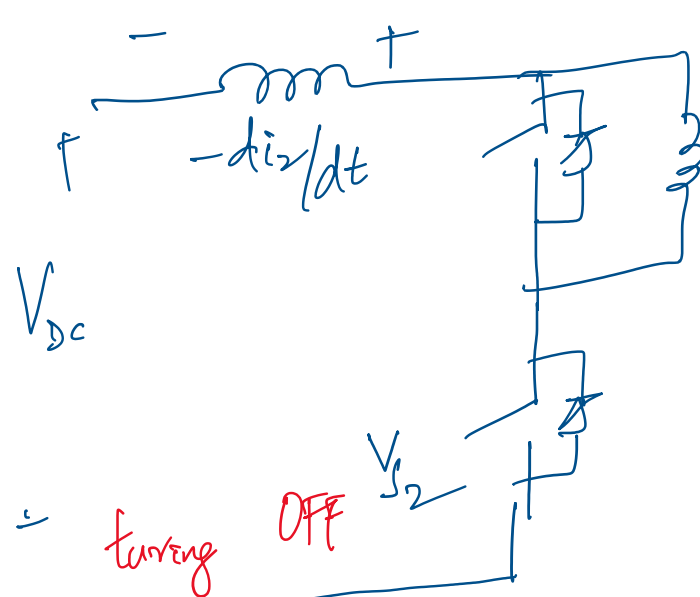
$$(1) \Rightarrow V_{DC} - L_{PCB} \frac{di_L}{dt} = V_{S2} \quad 800 - V_{dip}$$

known $\uparrow$ $\uparrow$ Cutq but dip ΔV



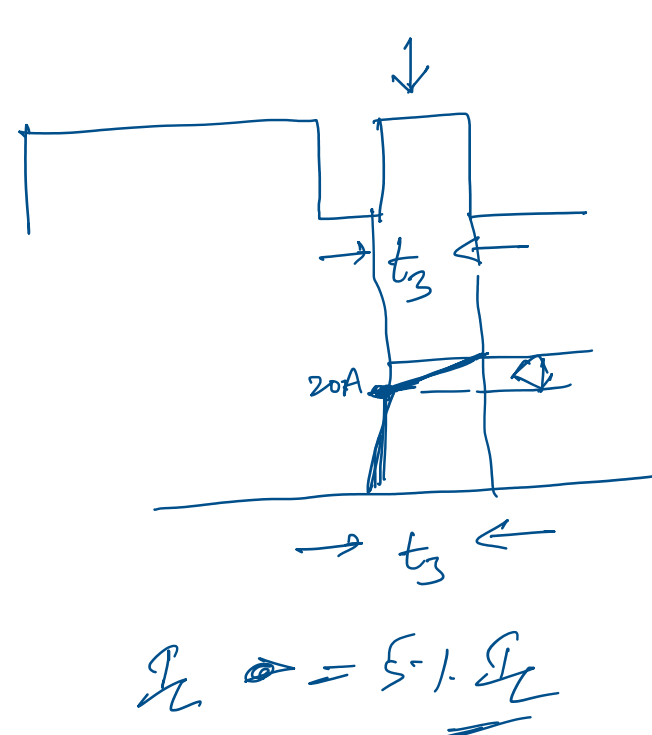
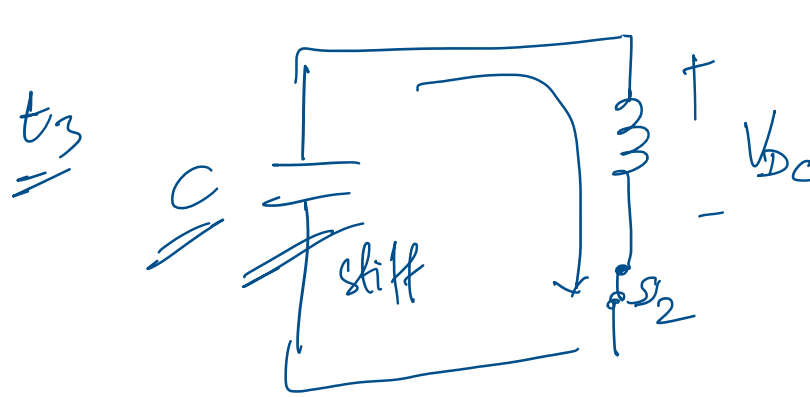
Overshoot across switch

$$V_{DC} + L_{PCB} \frac{di_L}{dt} = V_{S2}$$



Designing:- L, C

L : Turn on instant during t_3



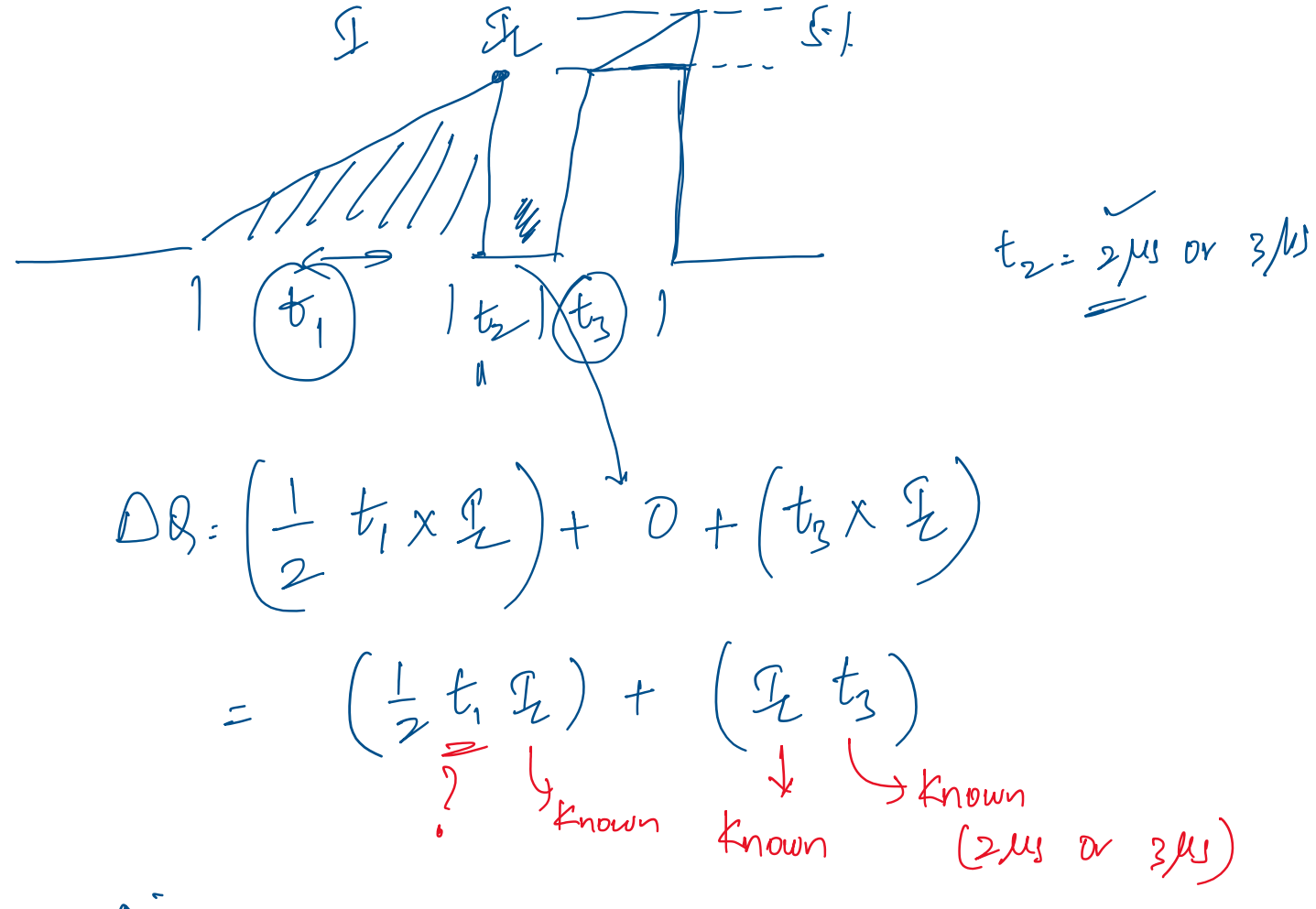
$$V = L \frac{di}{dt} \Rightarrow L = \frac{V \times t_3}{\Delta I} \quad t_3 = 2\mu s \text{ or } 3\mu s$$

$$\geq \frac{400 \times 2 \times 10^{-6}}{0.05 \times I_L} \geq L$$

$C := Q = CV$

$$C = \frac{Q}{\Delta V}$$

$\downarrow$ 1.1. dip



$$t_1: V = L \frac{di}{dt}$$

$$t_1 = \frac{L \times \Delta i}{V_{DC}} = \frac{L \times 20}{400} = \text{ } \mu s$$

CCL:

